MTH200 Chapter 5: Integration

5.3.1 The Fundamental Theorem Worksheet

1. Compute, if possible, the following definite integrals.

a) $\int_2^4 (4x-3)\,dx$

First we note that the function $4x-3$ is continuous on $[2,4]$, so we can apply FTCP2.

$$\int_2^4 (4x-3)\,dx = \left(2x^2-3x\right)\Big|_2^4 = \left[2(4)^2-3(4)\right]-\left[2(2)^2-3(2)\right] = [32-12]-[8-6] = 20-2 = 18$$

b) $\int_0^2 \left(6t^2+5e^t+2\right)dt$

First we note that the function $f(t)=6t^2+5e^t+2$ is continuous on $[0,2]$, so we can apply FTCP2.

$$\int_0^2 \left(6t^2+5e^t+2\right)dt = \left(2t^3+5e^t+2t\right)\Big|_0^2 = \left[2(2)^3+5e^2+2(2)\right]-\left[2(0)^3+5e^0+2(0)\right]$$

$$=\left[20+5e^2\right]-[5] = 15+5e^2$$

c) $\int_{-2}^{3} \frac{8(x-2)+x}{(x-2)}\,dx$

We note that $f(x)=\frac{8(x-2)+x}{(x-2)}$ is <u>*NOT*</u> continuous on $[-2,3]$, specifically because it is undefined at $x=2$. Therefore, the FTCP2 does not apply and we *currently* have no way to determine such an area. We cannot proceed.

d) $\int_1^4 \frac{18\sqrt{x}-x^3}{3x}\,dx$

First we note that the function is continuous on $[1,4]$, so we can apply FTCP2.

$$\int_1^4 \frac{18\sqrt{x}-x^3}{3x}\,dx = \int_1^4 \left(6x^{-1/2}-\frac{1}{3}x^2\right)dx = \left(12x^{1/2}-\frac{1}{9}x^3\right)\Big|_1^4 = \left[12\sqrt{4}-\frac{1}{9}(4)^3\right]-\left[12\sqrt{1}-\frac{1}{9}(1)^3\right]$$

$$=\left[24-\frac{64}{9}\right]-\left[12-\frac{1}{9}\right] = 24-12-\frac{64}{9}+\frac{1}{9} = 12-\frac{63}{9} = 12-7 = 5$$

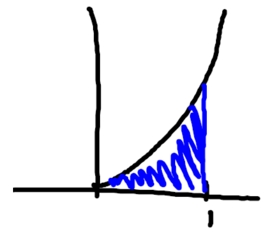
e) $\displaystyle\int_1^9 \frac{1}{2x}\,dx$

First we note that the function is continuous on $[1,9]$, so we can apply FTCP2.

$$\int_1^9 \frac{1}{2x}\,dx = \frac{1}{2}\int_1^9 \frac{1}{x}\,dx = \frac{1}{2}\ln|x|\Big|_1^4 = \frac{1}{2}\left[\ln(9)-\ln(1)\right] = \frac{1}{2}\ln(9) = \ln(3)$$

f) The area under the parabola $y = x^2$ from 0 to 1.

We note that $y = x^2$ is continuous on $[0,1]$, so by the FTCP2 we know that:



$$A = \int_0^1 x^2\,dx = \frac{1}{3}x^3\Big|_0^1 = \left[\frac{1}{3}(1)^3\right] - \left[\frac{1}{3}(0)^3\right] = \frac{1}{3} - 0 = \frac{1}{3}$$

2. What is wrong with the following calculation?

$$\int_0^2 \frac{-1}{(x-1)^2}\,dx = \frac{1}{(x-1)}\Bigg|_0^2 = \frac{1}{1} - \left(\frac{1}{-1}\right) = 1+1 = 2$$

The function is not continuous on $[0,2]$, specifically because it is undefined at $x=1$. Therefore, the FTCP2 does not apply and we *currently* have no way to determine such an area. We cannot proceed.

3. Suppose $w(x) = \displaystyle\int_5^{x^2} \frac{\ln(t+6)}{t}\,dt$. Find $w'(x)$.

$$w'(x) = \frac{\ln(x^2+6)}{x^2}\cdot 2x = \frac{2\ln(x^2+6)}{x}$$

4. Suppose $H(x) = \displaystyle\int_{\sqrt{x}}^{10} 6\left(t^4 - \frac{1}{t}\right)^{3/7} dt$. Find $H'(x)$.

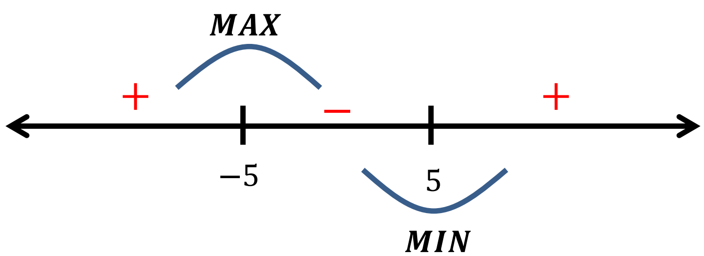
We note that $H(x) = \displaystyle\int_{\sqrt{x}}^{10} 6\left(t^4 - \frac{1}{t}\right)^{3/7} dt = -\int_{10}^{\sqrt{x}} 6\left(t^4 - \frac{1}{t}\right)^{3/7} dt$.

Therefore $H'(x) = -6\left(\left(\sqrt{x}\right)^4 - \dfrac{1}{\sqrt{x}}\right)^{3/7}\cdot\dfrac{1}{2\sqrt{x}} = -\dfrac{3}{\sqrt{x}}\left(x^2 - \dfrac{1}{\sqrt{x}}\right)^{3/7}$

5. Suppose $p(x)=\int_{8}^{x^3-75x}\frac{1}{t^4+100}dt$. Determine each interval upon which the graph of $p(x)$ is decreasing.

$$p'(x)=\frac{1}{\left(x^3-75x\right)^4+100}\cdot\left(3x^2-75\right)=\frac{3(x-5)(x+5)}{\left(x^3-75x\right)^4+100}$$

$p'(x)$ is never undefined.
$p'(x)=0$ when $x=\pm 5$.



$p(x)$ is decreasing on the interval:
$(-5,5)$

6. Find the average value of the function $f(x)=1+x^2$ on the interval $[-1,2]$.

$$f_{avg}=\frac{1}{2-(-1)}\int_{-1}^{2}\left(1+x^2\right)dx$$

Since $f(x)=1+x^2$ is continuous on $[-1,2]$ we know by the FTCP2 that:

$$\frac{1}{3}\int_{-1}^{2}\left(1+x^2\right)dx=\frac{1}{3}\left(x+\frac{x^3}{3}\right)\Bigg|_{-1}^{2}=\frac{1}{3}\left[2+\frac{(2)^3}{3}\right]-\frac{1}{3}\left[-1+\frac{(-1)^3}{3}\right]$$

$$=\frac{1}{3}\left[\frac{14}{3}\right]-\frac{1}{3}\left[-\frac{4}{3}\right]=\frac{14}{9}+\frac{4}{9}=\frac{18}{9}=2$$

7. Suppose $F(x)$ is a polynomial with $F'(x)=f(x)$. Given that:
$F(0)=2,\quad F(2)=8,\quad F(4)=28,\quad F(6)=68,\quad$ and $\quad F(8)=42$ find the average value of $f(x)$ on the interval $[2,6]$.

Since $F(x)$ is a polynomial then $f(x)$ is a polynomial, so $f(x)$ is integrable on the interval $[2,6]$.

$$f_{avg}=\frac{1}{6-2}\int_{2}^{6}f(x)dx=\frac{1}{4}\left[F(6)-F(2)\right]=\frac{1}{4}[68-8]=\frac{1}{4}[60]=15$$

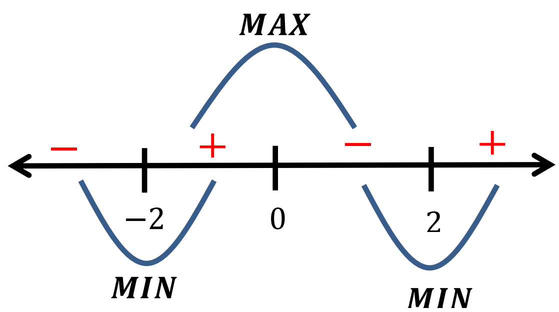
8. Suppose $w(x)=\int_{10}^{x^2}(t-4)(t+1)^6\,dt$. Determine all the intervals upon which $w(x)$ is increasing.

We note that $f(t)=(t-4)(t+1)^6$ is a continuous function from $[10,\infty)$, so we can apply FTCP1.

$$\begin{aligned} w'(x) &= (x^2-4)(x^2+1)^6(2x) \\ &= (x-2)(x+2)(x^2+1)^6(2x) \end{aligned}$$

(remember to use the chain rule)

Note that $w'(x)=0$ when $x=0$ *or* $x=\pm 2$.



$w(x)$ is increasing on the following intervals:

$$(-2,0)\cup(2,\infty)$$